Product Requirement Document

Product Name

Library Management System

Problem Statement

Manual library operations are often inefficient, prone to human error, and time-consuming, leading to a poor experience for both librarians and library members. This includes difficulties in tracking books, managing user accounts, recording issue and return dates, and accurately calculating overdue fines.

Goals

• Automate core library operations to improve efficiency.

• Reduce manual errors in book and user management.

• Enhance the user experience for library members and staff.

• Provide accurate and timely fine calculation and tracking.

• Improve accessibility of library resources information.

User Personas

• Librarian: Responsible for managing books, users, and all library operations. Needs an efficient system to perform daily tasks.

• Library Member: Wants to easily borrow, return, and renew books. Needs to view their borrowing history and outstanding fines.

• Administrator: Manages system settings, user roles, and generates reports.

Functional Requirements

• Ability to add, edit, and delete book records (title, author, ISBN, genre, etc.).

• Ability to register, update, and delete library member accounts (name, address, contact, membership ID).

• Ability to record book issuance to members, including issue date and due date.

• Ability to record book returns, including return date.

• Automated calculation of overdue fines based on predefined rules.

• Ability to search for books by various criteria (title, author, ISBN).

• Ability to search for library members by ID or name.

• Ability to view a member's borrowing history and current borrowed books.

• Ability to generate reports on borrowed books, overdue books, and fine collections.

• User authentication and authorization for different roles.

Non Functional Requirements

• Performance: The system should respond to user requests within 2 seconds for common operations.

• Security: User data and book information must be protected against unauthorized access. Authentication and authorization mechanisms must be robust.

• Usability: The system should have an intuitive and easy-to-use interface for all user types.

• Reliability: The system should be available 99.5% of the time during operational hours.

• Scalability: The system should be able to handle an increasing number of books and users without significant performance degradation.

• Maintainability: The code should be well-documented and easy to maintain and update.

Constraints

• The system must be developed using open-source technologies.

• Deployment will be on a cloud-based infrastructure.

• Integration with existing payment gateways for fine collection is not required in the initial phase.

• Project completion deadline is 6 months from the start date.

Assumptions

• Library staff will receive adequate training on how to use the new system.

• Reliable internet connectivity will be available for system users.

• Hardware infrastructure will meet the minimum requirements for system deployment.

• All necessary data for initial setup (existing books, members) will be provided in a structured format.

Out Of Scope

• E-book management and digital content distribution.

• Integration with external payment systems for fine collection.

• Inventory management for non-book library items (e.g., DVDs, magazines).

• Public-facing online catalog for searching without login.

• Advanced analytics and predictive modeling for book recommendations.